

Construct a deterministic TM of the kind you prefer, which decides the following language:

$$\Sigma = \{w \in \{0,1\}^* \mid w \text{ contains at least an occurrence of } 011 \text{ and no occurrence of } 100\}.$$

Study the complexity of the TM you have defined.

The concept classes of axis-aligned squares in $\mathbb{R}^2$ is:

Scegli una o più alternative:

- ☐ PAC learnable, but not efficiently so.
- ☐ Not even PAC learnable
- ☒ PAC learnable ✓
- ☒ Efficiently PAC learnable. ✓

Your answer is correct.

Le risposte corrette sono: PAC learnable, Efficiently PAC learnable.

A problem $\square$ is said to be **EXP**-complete if it is in **EXP** and any problem in **EXP** can be reduced, in polynomial time, to $\square$. Suppose a friend of yours comes to you claiming that he found an **EXP**-complete problem which is also in **NP**. What can you say?

Scegli una o più alternative:

- ☒ That your friend is perhaps right, but that, if right, he would have proved $\mathbf{P} \neq \mathbf{NP}$. ✓
- ☐ That your friend is certainly wrong, because such a problem cannot exist.
- ☐ That your friend is certainly right, because finding such a problem is very easy.
- ☐ That your friend is perhaps right, but that, if right, he would have proved $\mathbf{P} = \mathbf{NP}$.

Your answer is correct.

La risposta corretta è: That your friend is perhaps right, but that, if right, he would have proved $\mathbf{P} \neq \mathbf{NP}$.

The problem of deciding whether an undirected graph G contains a clique of size equal to the size of G is:

Scegli una o più alternative:

- ☐ Undecidable, and **NP**-hard.
- ☒ In the class **EXP** ✓
- ☐ Decidable in polynomial time
- ☐ Not in the class **NP**

Your answer is partially correct.

Hai selezionato correttamente 1.

Le risposte corrette sono: Decidable in polynomial time, In the class **EXP**

The problem of deciding whether an undirected graph G contains a clique of size equal to the size of G is:

Scegli una o più alternative:

- ☐ Undecidable, and **NP**-hard.
- ☒ In the class **EXP** ✓
- ☐ Decidable in polynomial time
- ☐ Not in the class **NP**

Which of the following sets can be easily encoded as binary strings?

Scegli una o più alternative:

- ☒ Strings on an alphabet of size 10^2 ✓
- ☒ Strings on an alphabet of size 10^4 ✓
- ☐ Vectors of complex numbers
- ☒ Vectors of rational numbers ✓

Your answer is correct.

Le risposte corrette sono: Strings on an alphabet of size 10^2 , Strings on an alphabet of size 10^4 , Vectors of rational numbers

Let f, g be the functions defined as $f(n) = 2^{-30} n^{2^n} \log^5 n$ and $g(n) = 2^{10} \frac{n^{3^n}}{\log^2 n}$.

Scegli una o più alternative:

- ☐ $f \in \Omega(g)$
- ☒ $f \in O(g)$ ✓
- ☐ $f \in \Theta(g)$

Your answer is correct.

La risposta corretta è: $f \in O(g)$

Consider the problem

$3SATVAR = \{ \langle F \rangle \mid F \text{ is a satisfiable 3CNF in which at most 4 distinct boolean variables occur.} \}$

What's the complexity of 3SATVAR?

You are required to prove that the following problem $\square$ is in **NP**. To do that, you can give a TM or define some pseudocode. The language $\square$ includes precisely those binary strings which are encodings of triples in the form $(\mathbb{G}, n, m)$ where $\mathbb{G}$ is an undirected graph, n, m are natural numbers, and $\mathbb{G}$ contains two independent sets of size at least equal to n and m , respectively. (Two sets are disjoint if their intersection is empty.) Can you say anything else about the complexity of $\square$?